

Problem statement

Image segmentation is an important application of image processing techniques in which some, or all, objects in an image are isolated from the background. This can be achieved by using the Chan-Vese algorithm for active contours, a flexible method which aims to identify and separate the mean intensity of different image regions. However, this method seems insufficient in images where the mean intensity of the regions is not significantly different. In this project, we aim to characterize the foreground and the background of an image using distribution of intensities or other features to solve more challenging situations.

Choice of method

To overcome the limitations of the Chan-Vese mean-intensities-based approach, we follow the probabilistic framework presented by Dahl and Dahl (2017). This framework assumes that the foreground and background regions present different distributions of image features rather than work directly on image intensities. In that way, we can deform the curve such that the differences between distributions increases. We have tested two approaches based in this framework:

- **Intensity-based distributions:** The normalized histograms of the inner and outer areas to the curve are used to represent the difference between distributions. This approach allow us to differentiate regions with different intensity distributions.
- **Patch-based distributions:** The use of image patches (sub-regions of the images) might be a better descriptor in the case of textured images, in which the intensity distributions of the background and foreground might overlap.

The figure 1a exemplifies a suitable case for the intensity-based approach, while the 1b is an example of a suitable patch-based approach.

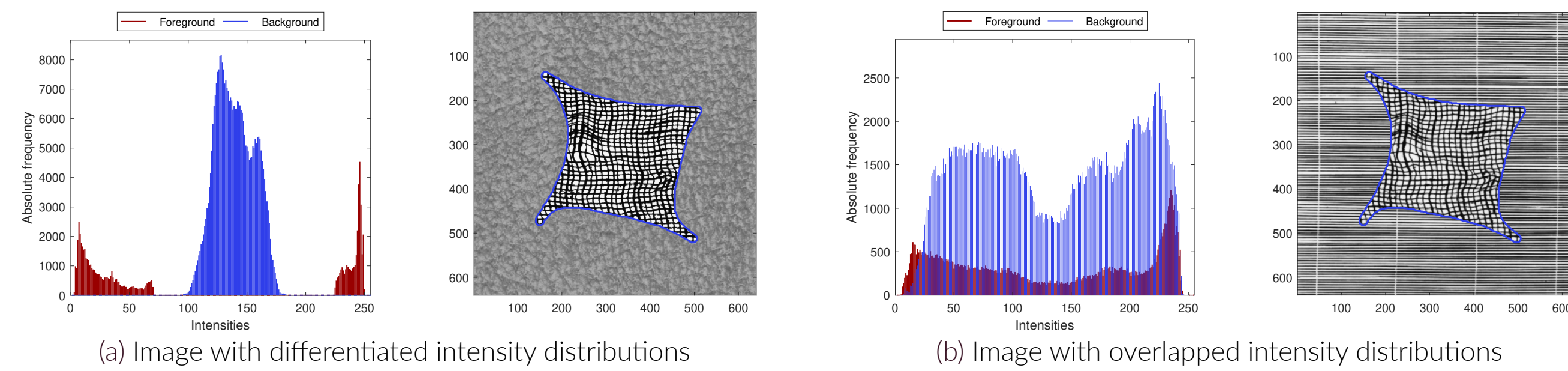


Figure 1. Feasible segmentation problems for the intensity-based and patch-based approaches, respectively

The algorithm

We consider the image I , which is divided by an arbitrary curve, C , into an inner and outer region. Therefore, we can extract the histograms of the inner and outer regions and dividing those by their respective areas to obtain normalized frequencies, $f_{in}(k)$, and $f_{out}(k)$. If we normalize those values as

$$p_{in}(k) = \frac{1}{f_{in}(k) + f_{out}(k)} f_{in}(k)$$

where k represents the pixel intensity, we can consider them as empirical values of the probability of that intensity to belong to the inner distribution. Finally, we construct a probability image in which the pixels of the image I with value k are substituted by $p_{in}(k)$. P_{in} is updated in each iteration of the image, and acts as indicator of foreground-background segmentation for evolving the curve. The employed external forces are:

$$\vec{F}_{ext} = (P_{in} - P_{out})\vec{N}$$

Patch-based probabilities

We start extracting M -by- M patches from the image I , those patches are grouped through k-means clustering. The obtained centroids are called *dictionary*. Then, image patches of the whole image are assigned to the nearest dictionary element. The representation of each pixel with the value of its assigned cluster is called *assignment image*, S .

Finally, similar to the intensity-approach, we estimate the histograms of the outer and inner regions. But in this case, each pixel contributes to the histogram with the intensity values grouped in its same cluster.

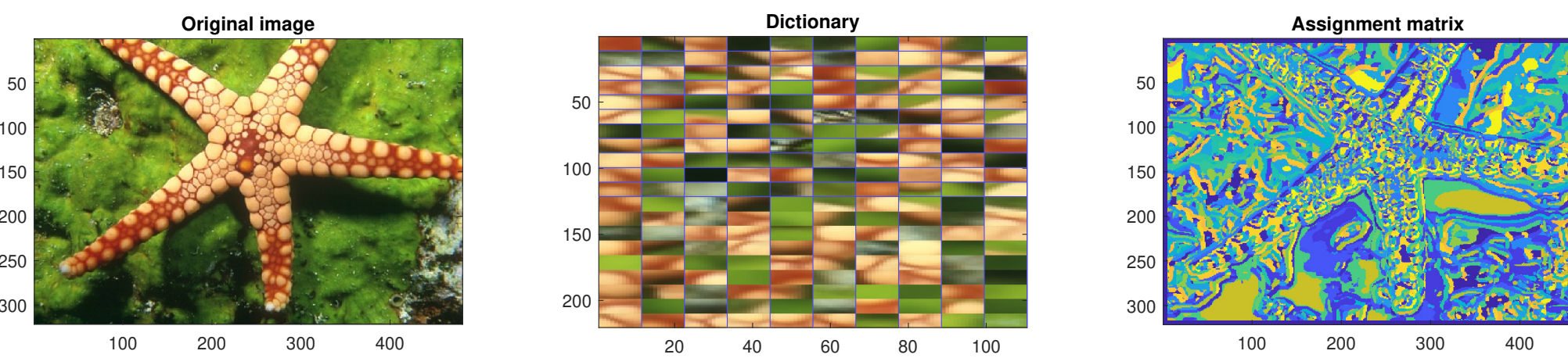


Figure 2. Example of dictionary with 200 clusters and the assignment image

Implementation

The iterative part of the method, the estimation of P_{in} , can be implemented in terms of linear mapping, which makes the method computationally reasonable. The general steps use an square bullet, the exclusive steps for the patch approach use an asterisk bullet:

- * Extract image patches of size M -by- M from the image, and order them column-wise.
- * Perform k-means clustering, and assign to every patch a dictionary entry. (Note: The patches might (or not) be normalized to unit length).
- * Reorder the patches to obtain the assignment image, S .
- Obtain the biadjency matrix, B :
 - Get a matrix in which the rows represents the pixel position and the columns the intensity/assigned cluster of that pixel.
 - * Get a matrix in which rows represents the cluster number and the columns the intensities in the patches of that cluster.
 - * Compute the matrix product, and divide by the sum of rows from B .
- Obtain the probability image for each iteration:
 - Obtain the normalized histograms as: $f_{in} = \frac{B^T c_{in}}{A_{in}}$
 - Element-wise normalization to obtain p_{in} .
 - $P_{in} = B p_{in}$.
 - Reshape P_{in} into the image grid.
- Deform the curve

Results

We have tested the different implementations providing as initial curve a small circle that includes foreground and background regions of the image. Firstly, the two approaches are compared with the textured image, in which we observe how the patch approach converges into a better solution than the intensity-distribution based approach.

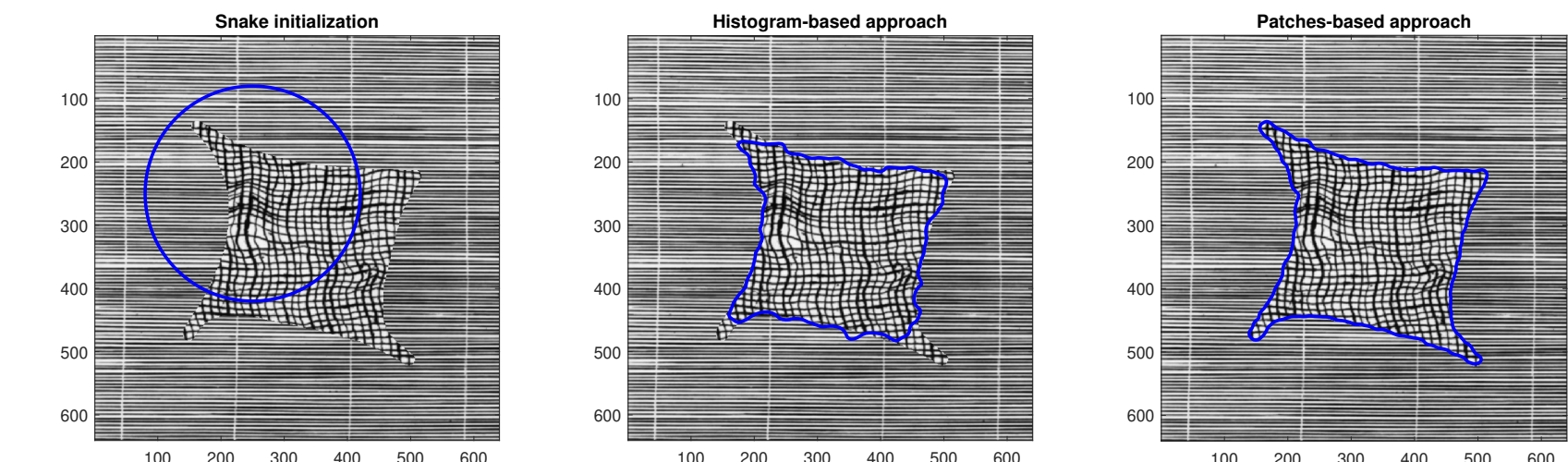


Figure 3. Comparison of approaches in a synthetic image

Then, the patch approach has been tested in different natural RGB images. Figure 4 shows the segmentation and the P_{in} image, in which we see how the probability of *being inside* of the flower was higher in the last iteration.

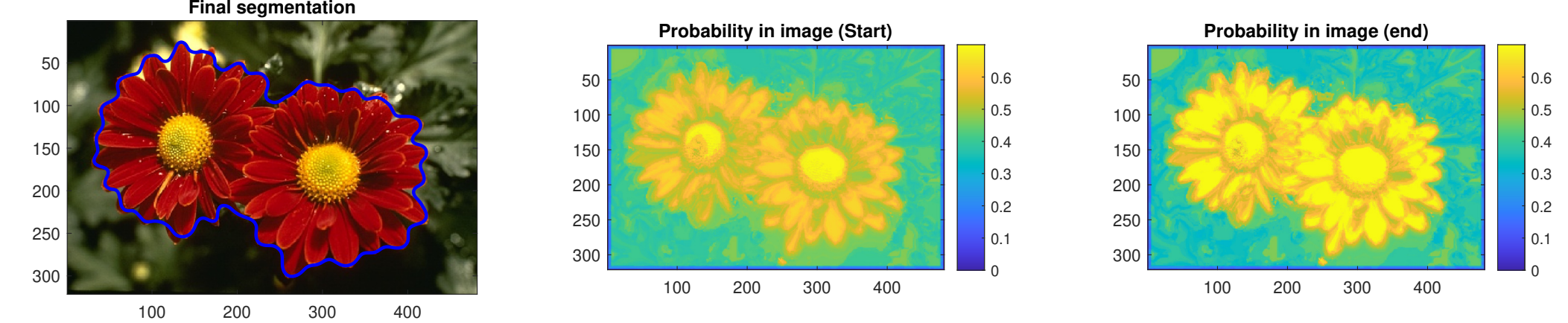


Figure 4. Segmentation and evolution of the P_{in} image. The curve was initialized as a small circle in the middle of the flower.

We realised that the use of small patches implied to converge in a local minimum solution, while too large patches implied a worse localization of the segmentation boundary. We found that in some cases averaging the probability images obtained from different patch sizes might lead to better boundary localization:

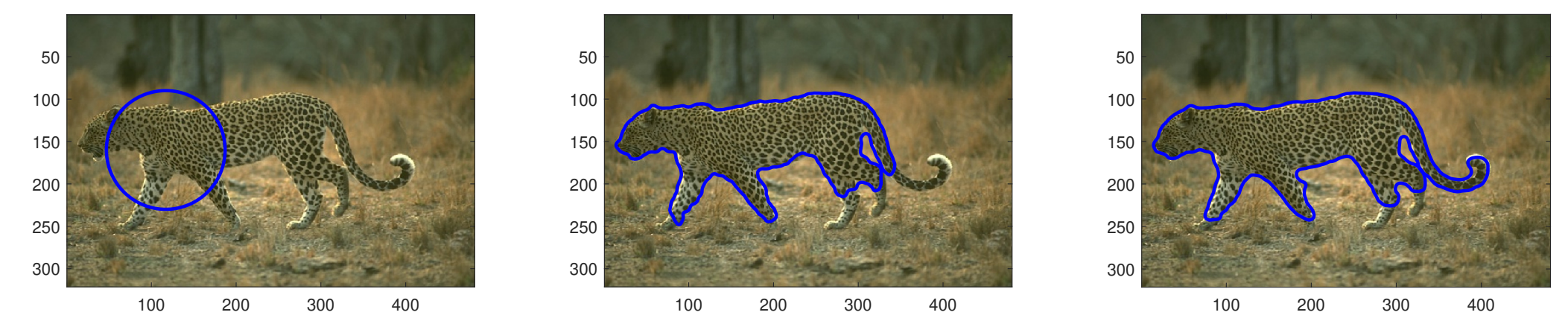


Figure 5. Initialization of the curve, segmentation with patches of 7-by-7 and segmentation averaging the P_{in} of patches of 5-by-5 and 11-by-11

Conclusions

The probabilistic framework provides a simple and efficient extension of the original Chan-Vese algorithm, which permits to expand the use of deformable models to more general situations. The results are generally acceptable with a relative weak initialization, but a correct selection of the parameters is necessary to get the expected results.